Funcții olomorfe

Def Fc. complexă

Fie $A \subset \mathbb{C}$ o mulțime. Funcția $f: A \to \mathbb{C}$ se numește fc. complexă

Fie $z \in A$, $z = x + iy$ cu $f(z) = w \in \mathbb{C}$ $\quad x, y, u, v \in \mathbb{R}$

$w = u + iv$

$f = P + iQ$ unde $P = \operatorname{Re} f$ $\quad P, Q : A \subset \mathbb{R}^2 \longrightarrow \mathbb{R}$

$Q = \operatorname{Im} f$ $\quad$ fc. de var. reale

egalitatea $f(z) = w \iff u = P(x,y)$

$v = Q(x,y)$

Def Fc. continuă

Fie $A \subset \mathbb{C}$ mulțime deschisă

$z_0 \in A$, $z_0 = x_0 + iy_0$ $\quad f: A \longrightarrow \mathbb{C}$, $f(z) = P + iQ$

Fc. f este continuă în punctul z_0 dacă și numai dacă funcțiile P, Q sunt simultan continue în (x_0, y_0)

f cont. dacă $\lim\limits_{z \to z_0} f(z) = f(z_0)$

Def Fc. $\mathbb{C}$ derivabilă

Fie $A \subset \mathbb{C}$ mulțime deschisă

$z_0 \in A$, $z_0 = x_0 + iy_0$ $f: A \longrightarrow \mathbb{C}$, $f(z) = P + iQ$

Fc. f se numește $\mathbb{C}$ derivabilă în punctul z_0 dacă

$$\exists \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = l \in \mathbb{C}$$

$l = f'(z_0)$ derivata complexă a funcției f în pct. z_0

Def Fc. olomorfă

Fc. f se numește olomorfă dacă f este definită și $\mathbb{C}$ derivabilă în toate punctele mulțimii A.

Def Fc. olomorfă în z_0

Fc. f se numește olomorfă în $z_0 \in A$ dacă f este olomorfă pe o vecinătate a punctului z_0

Fie f olomorfă în z_0 atunci

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} \quad \text{pt. } h \in \mathbb{C}$$

În particular pentru $h \in \mathbb{R}$

$$f'(z_0) = \lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{h \to 0} \frac{f(z_0 + ih) - f(z_0)}{ih}$$

Dem

$$\lim_{h\to 0}\frac{P(x_0+h,y_0)+iQ(x_0+h,y_0)-P(x_0,y_0)-iQ(x_0,y_0)}{h}$$

$$=\lim_{h\to 0}\frac{P(x_0,y_0+h)+iQ(x_0,y_0+h)-P(x_0,y_0)-iQ(x_0,y_0)}{ih}$$

$$\Rightarrow \lim_{h\to 0}\frac{P(x_0+h,y_0)-P(x_0,y_0)}{h}+i\lim_{h\to 0}\frac{Q(x_0+h,y_0)-Q(x_0,y_0)}{h}$$

$$=\lim_{h\to 0}\frac{Q(x_0,y_0+h)-Q(x_0,y_0)}{h}-i\lim_{h\to 0}\frac{P(x_0,y_0+h)-P(x_0,y_0)}{h}$$

Pp că funcțiile P, Q au derivate parțiale în (x_0,y_0) în raport cu x, y.

$$\Rightarrow \begin{cases}\frac{\partial P}{\partial x}(x_0,y_0)=\frac{\partial Q}{\partial y}(x_0,y_0)\\ \frac{\partial P}{\partial y}(x_0,y_0)=-\frac{\partial Q}{\partial x}(x_0,y_0)\end{cases}$$

relațiile Cauchy – Riemann (C-R) în punctul (x_0,y_0)

Teorema 1

Fie $A\subset\mathbb{C}$ deschisă, $z_0\in A$, $z_0=x_0+iy_0$

$f: A\to\mathbb{C}$, $f=P+iQ$

Atunci f este olomorfă în $z_0 \Leftrightarrow P, Q$ sunt fc diferențiabile în (x_0,y_0) și au loc cond. C-R în (x_0,y_0)

Dem

„$\Rightarrow$" f olomorfă în $z_0 \Rightarrow \exists \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0} = f'(z_0) \in \mathbb{C}$ // not $c = a+ib$

Fie $\beta(z) = \begin{cases} \frac{f(z)-f(z_0)}{z-z_0} - f'(z_0), & z \neq z_0 \\ 0, & z = z_0 \end{cases}$ $\qquad \lim_{z\to z_0} \beta = 0$

Fie $\alpha(z) = \begin{cases} \beta(z) \cdot \frac{z-z_0}{|z-z_0|}, & z \neq z_0 \\ 0, & z = z_0 \end{cases}$

$$\lim_{z\to z_0} |\alpha(z)| = \lim_{z\to z_0} |\beta(z)| = 0 \Rightarrow \lim_{z\to z_0} \alpha(z) = 0 \quad \underline{I}$$

$$f(z) - f(z_0) = \alpha(z)|z-z_0| + c(z-z_0) \quad \underline{II}$$

$$\alpha(z) = \mathrm{Re}\,\alpha + i\,\mathrm{Im}\,\alpha \xrightarrow{I} \lim_{(x,y)\to(x_0,y_0)} \mathrm{Re}\,\alpha(x,y) = 0$$

$$\lim_{(x,y)\to(x_0,y_0)} \mathrm{Im}\,\alpha(x,y) = 0 \quad \underline{III}$$

$\underline{II}$ devine $P(x,y) + iQ(x,y) - P(x_0,y_0) - iQ(x_0,y_0) = (\mathrm{Re}\,\alpha + i\,\mathrm{Im}\,\alpha)\sqrt{(x-x_0)^2+(y-y_0)^2} + (a+bi)(x+iy-x_0-iy_0) \Rightarrow$

$$\Rightarrow \begin{cases} P(x,y) - P(x_0,y_0) = \mathrm{Re}\,\alpha \sqrt{(x-x_0)^2+(y-y_0)^2} + a(x-x_0) - \\ \qquad\qquad - b(y-y_0) \\ Q(x,y) - Q(x_0,y_0) = \mathrm{Im}\,\alpha \sqrt{(x-x_0)^2+(y-y_0)^2} + a(y-y_0) + \\ \qquad\qquad + b(x-x_0) \end{cases} \Rightarrow$$

$\xrightarrow[\text{def. dif.}]{\text{f III}}$ P, Q diferențiabile în $(x_0,y_0) \Rightarrow$

(sem I)

$$\Rightarrow \begin{cases} a = \dfrac{\partial P}{\partial x}(x_0,y_0) & b = \dfrac{\partial Q}{\partial x}(x_0,y_0) \\ -b = \dfrac{\partial P}{\partial y}(x_0,y_0) & a = \dfrac{\partial Q}{\partial y}(x_0,y_0) \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \dfrac{\partial P}{\partial x}(x_0,y_0) = \dfrac{\partial Q}{\partial y}(x_0,y_0) \\ \dfrac{\partial P}{\partial y}(x_0,y_0) = -\dfrac{\partial Q}{\partial x}(x_0,y_0) \end{cases}$$ C-R (condiția Cauchy-Riemann)

„$\Leftarrow$" P, Q diferențiabile în $(x_0,y_0) \xrightarrow[\text{dif}]{\text{def.}}$

$$P(x,y) - P(x_0,y_0) = \alpha_1(x,y)\sqrt{(x-x_0)^2+(y-y_0)^2} + \frac{\partial P}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial P}{\partial y}(x_0,y_0)(y-y_0) \quad \text{IV}$$

$$Q(x,y) - Q(x_0,y_0) = \alpha_2(x,y)\sqrt{(x-x_0)^2+(y-y_0)^2} + \frac{\partial Q}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial Q}{\partial y}(x_0,y_0)(y-y_0) \quad \text{V}$$

unde $\lim\limits_{(x,y)\to(x_0,y_0)} \alpha_1(x,y) = 0$

$\lim\limits_{(x,y)\to(x_0,y_0)} \alpha_2(x,y) = 0$

C-R: $\begin{cases} \dfrac{\partial P}{\partial x}(x_0,y_0) = \dfrac{\partial Q}{\partial y}(x_0,y_0) \overset{\text{not}}{=} a \\ \dfrac{\partial P}{\partial y}(x_0,y_0) = -\dfrac{\partial Q}{\partial x}(x_0,y_0) \overset{\text{not}}{=} b \end{cases}$

Notăm $\alpha(z) = \alpha_1 + i\alpha_2$

$\overline{IV} + \overline{V} \Rightarrow f(z) - f(z_0) = \alpha(z)|z-z_0| + (a+ib)(z-z_0)$

$\Rightarrow \dfrac{f(z)-f(z_0)}{z-z_0} = \alpha(z)\dfrac{|z-z_0|}{z-z_0} + (a+ib)$ $\quad$ not c

$\lim\limits_{z\to z_0} \alpha(z) = 0$

$\not\Rightarrow \dfrac{f(z)-f(z_0)}{z-z_0} = c \in \mathbb{C}$

$\Rightarrow f$ olomorfă

Corolar 1 Fie $A \subset \mathbb{C}$ deschisă, $f: A \to \mathbb{C}$, $f = P + iQ$
Dacă $P, Q \in C^1(A)$ și au loc condițiile C-R
atunci fc. f olomorfă pe A

Dem $P, Q \in C^1 \xrightarrow[\text{semi}]{\text{th}} P, Q$ dif. $\xrightarrow[\text{au loc cond C-R}]{\text{th I}}$ f olomorfă

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right)$$

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right)$$

derivată aleorală

Corolar 2 Fie $A \subset \mathbb{C}$ deschisă, $f: A \to \mathbb{C}$, $f = P + iQ$
Dacă $P, Q \in C^1(A)$ și $\dfrac{\partial f}{\partial \bar{z}} = 0$ atunci f olomorfă

Dem $\dfrac{\partial f}{\partial \bar{z}} = 0 \Rightarrow \dfrac{1}{2}\left[\dfrac{\partial P}{\partial x} + i\dfrac{\partial Q}{\partial x} + i\left(\dfrac{\partial P}{\partial y} + i\dfrac{\partial Q}{\partial y}\right)\right] = 0 \Rightarrow$

$$\Rightarrow \begin{cases} \dfrac{\partial P}{\partial x} - \dfrac{\partial Q}{\partial y} = 0 \\ \dfrac{\partial Q}{\partial x} + \dfrac{\partial P}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} \dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y} \\ \dfrac{\partial P}{\partial y} = -\dfrac{\partial Q}{\partial x} \end{cases}$$

$P, Q \in C^1$ $\quad$ C-R $\xrightarrow{\text{cor 1}}$ f olomorfă

Def Fie $A \subset \mathbb{R}^2$ deschisă; $u: A \to \mathbb{R}$ de clasă $C^2(A)$

Funcția u se numește armonică dacă

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Corolar 3 Fie $A \subset \mathbb{C}$ deschisă, $f: A \to \mathbb{C}$, $f = P + iQ$

$P, Q \in C^2(A)$ f olomorfă atunci P și Q sunt funcții armonice

Dem $\Delta P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial P}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial P}{\partial y}\right)$

$= \frac{\partial}{\partial x}\left(\frac{\partial Q}{\partial y}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial Q}{\partial x}\right)$

f olomorfă $\overset{th.}{\Rightarrow}$ $\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases}$ $\Rightarrow \Delta P = \frac{\partial^2 Q}{\partial x \partial y} - \frac{\partial^2 Q}{\partial y \partial x}$

$\overset{th.\ Schw.}{\Rightarrow} \Delta P = 0$

$\Rightarrow P$ funcție armonică

Analog, pt Q $\Delta Q = \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2} = \frac{\partial}{\partial x}\left(\frac{\partial Q}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial Q}{\partial y}\right)$

$= \frac{\partial}{\partial x}\left(-\frac{\partial P}{\partial y}\right) + \frac{\partial}{\partial y}\left(\frac{\partial P}{\partial x}\right)$

f olomorfă $\overset{th.}{\Rightarrow}$ $\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases}$ $\Rightarrow$

$\Rightarrow \Delta Q = -\frac{\partial^2 P}{\partial x \partial y} + \frac{\partial^2 P}{\partial y \partial x} \overset{th.\ Schw}{=} 0 \Rightarrow Q$ funcție armonică

① Să se studieze dacă următoarele funcții sunt olomorfe și să se afle punctele în care acestea sunt (dacă sunt olomorfe)

a) $f(z) = z^2 + e^{iz}$

$$f(x+iy) = (x+iy)^2 + e^{i(x+iy)} = x^2 + 2ixy - y^2 + e^{ix-y}$$

$$= x^2 + 2ixy - y^2 + e^{-y}(\cos x + i \sin x)$$

$$= \underbrace{x^2 - y^2 + \cos x e^{-y}}_{P} + i\underbrace{(2xy + e^{-y}\sin x)}_{Q}$$

$P, Q \in C^1$

$$\frac{\partial P}{\partial x} = 2x - e^{-y}\cdot \sin x \quad \Rightarrow \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$$

$$\frac{\partial Q}{\partial y} = 2x - e^{-y}\sin x$$

$$\frac{\partial P}{\partial y} = -2y - e^{-y}\cos x \quad \Rightarrow \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x}$$

$$\frac{\partial Q}{\partial x} = 2y + e^{-y}\cos x$$

$\xrightarrow[\text{Cor 1}]{\text{C-R}}$ f olom. pe $\mathbb{R}$

b) $f(z) = \text{Re}(z^2) - i\,\text{Im}(z^2)$

$$f(x+iy) = \text{Re}(x^2 + 2ixy - y^2) - i\,\text{Im}(x^2 + 2ixy - y^2)$$

$$= \underbrace{x^2 - y^2}_{P} - \underbrace{2ixy}_{Q}$$

$P, Q \in C^1$

$$\frac{\partial P}{\partial x} = 2x \qquad \frac{\partial Q}{\partial y} = -2x$$

$$\frac{\partial P}{\partial y} = -2y \qquad \frac{\partial Q}{\partial x} = -2y$$

Cond. C-R $\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \Rightarrow 2x = -2x \Rightarrow x = 0 \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \Rightarrow -2y = 2y \Rightarrow y = 0 \end{cases} \Rightarrow z = 0$

$\Rightarrow f$ olomorfă în $z = 0$

Teorema 2 fie $D \subset \mathbb{R}^2$ domeniu simplu conex
$P: D \to \mathbb{R}$ o funcție armonică atunci
$\exists Q: D \to \mathbb{R}$ armonică a.î.
$f: D \to \mathbb{C}$, $f = P + iQ$ este olomorfă pe D

Dem Dacă ar exista Q a.î. $f = P + iQ$ olomorfă atunci (cf th 1) ar fi îndeplinite condițiile C-R

$$\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$$

$$\frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x}$$

$$dQ = \frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy = -\frac{\partial P}{\partial y}dx + \frac{\partial P}{\partial x}dy$$

Considerăm forma diferențială $\omega = \underbrace{-\frac{\partial P}{\partial y}}_{P_1}dx + \underbrace{\frac{\partial P}{\partial x}}_{Q_1}dy$

$$\frac{\partial P_1}{\partial y} = \frac{\partial}{\partial y}\left(\frac{-\partial P}{\partial y}\right) = \frac{-\partial^2 P}{\partial y^2}$$

$$\frac{\partial Q_1}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial P}{\partial x}\right) = \frac{\partial^2 P}{\partial x^2}$$

P armonică $\Rightarrow \Delta P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} = 0 \Rightarrow \frac{\partial^2 P}{\partial y^2} = -\frac{\partial^2 P}{\partial x^2}$

$$\Rightarrow -\frac{\partial P_1}{\partial y} = -\frac{\partial Q_1}{\partial x} \Rightarrow \frac{\partial P_1}{\partial y} = \frac{\partial Q_1}{\partial x} \Rightarrow$$

$\Rightarrow \omega$ închisă

~~$\Rightarrow$~~ dar D domeniu simplu conex

$\Rightarrow \omega$ exactă $\Rightarrow \exists Q \in C^2$ a.î. $dQ = \omega \Rightarrow$

$$\Rightarrow \frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy = -\frac{\partial P}{\partial y}dx + \frac{\partial P}{\partial x}dy \Rightarrow \begin{cases} \frac{\partial Q}{\partial x} = -\frac{\partial P}{\partial y} \\ \frac{\partial Q}{\partial y} = \frac{\partial P}{\partial x} \end{cases} \text{C-R}$$

$\overset{\text{Cor 1}}{\Longrightarrow} f$ olomorfă

② Să se determine funcția olomorfă $f: \mathbb{C} \to \mathbb{C}$

$f = P + iQ$

a) $Q(x,y) = e^x \sin y + \frac{y}{x^2+y^2}$; $f(1) = 1$

P1 Verificăm că f_c. armonică ($\Delta Q = 0$)

$$\Delta Q = \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2}$$

$$\frac{\partial Q}{\partial x} = e^x \sin y + \frac{-y \cdot 2x}{(x^2+y^2)^2}$$

$$\frac{\partial^2 Q}{\partial x^2} = e^x \sin y - \frac{2y(x^2+y^2)^2 - 2xy \cdot 2(x^2+y^2) \cdot 2x}{(x^2+y^2)^{4\,3}}$$

$$= e^x \sin y - 2 \cdot \frac{x^2y + y^3 - 4x^2y}{(x^2+y^2)^3} = e^x \sin y - 2 \cdot \frac{y^3 - 3x^2y}{(x^2+y^2)^3}$$

$$\frac{\partial Q}{\partial y} = e^x \cos y + \frac{x^2+y^2 - y \cdot 2y}{(x^2+y^2)^2} = e^x \cos y + \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 Q}{\partial y^2} = e^x(-\sin y) + \frac{-2y(x^2+y^2)^2 - (x^2-y^2)2(x^2+y^2) \cdot 2y}{(x^2+y^2)^{4\,3}}$$

$$= e^x(-\sin y) - 2\,\frac{x^2y + y^3 + 2x^2y - 2y^3}{(x^2+y^2)^3} = -e^x \sin y - \frac{2x^2y - y^3}{(x^2+y^2)^3}$$

$$\frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2} = 0 \Rightarrow \Delta Q = 0 \Rightarrow Q \text{ armonică}$$

P2 Aflăm P a.î. $f = P + iQ$

$$\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = e^x \cos y + \frac{x^2+y^2 - y \cdot 2y}{(x^2+y^2)^2} = e^x \cos y + \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$\frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = -\left[e^x \sin y + \frac{-y \cdot 2x}{(x^2+y^2)^2}\right] = -e^x \sin y + \frac{2xy}{(x^2+y^2)^2} \quad \text{II}$$

$$\text{II} \xrightarrow{\int dy} P(x,y) = \int -e^x \sin y + \frac{2xy}{(x^2+y^2)^2}\, dy$$

$$P(x,y) = e^x\cos y - \frac{x}{x^2+y^2} + C(x) \xrightarrow{\text{deriv } x}$$

$$\frac{\partial P}{\partial x} = e^x\cos y - \frac{x^2+y^2 - x\cdot 2x}{(x^2+y^2)^2} + C'(x) = e^x\cos y + \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\Rightarrow C'(x) = 0 \Rightarrow C(x) = k,\ k \in \mathbb{R}$$

$$P(x,y) = e^x\cos y - \frac{x}{x^2+y^2} + k,\ k \in \mathbb{R}$$

$$f = P + iQ = e^x\cos y - \frac{x}{x^2+y^2} + k + i\left(e^x\sin y + \frac{y}{x^2+y^2}\right)$$

$$f = e^x\underbrace{(\cos y + i\sin y)}_{=e^{iy}} - \frac{x-iy}{x^2+y^2} + k = e^{x+iy} - \frac{x-iy}{x^2+y^2} + k$$

$$f(z) = e^z - \frac{\overline{z}}{|z|^2} + k = e^z - \frac{1}{z} + k$$

$$f(1) = 1 \Rightarrow e - 1 + k = 1 \Rightarrow k = 2 - e$$

$$\Rightarrow f(z) = e^z - \frac{1}{z} + 2 - e$$

b) $P(x,y) = \underbrace{f(x^2-y^2)}_{=\alpha(x,y)}$, $f \in C^2$

$$\Delta P = 0 \Rightarrow \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} = 0$$

$$\frac{\partial P}{\partial x} = f'(\alpha)\cdot\frac{\partial \alpha}{\partial x} = f'(\alpha)\cdot 2x$$

$$\frac{\partial^2 P}{\partial x^2} = 2\cdot\left[f''(\alpha)\cdot 2x\cdot x + f'(\alpha)\cdot 1\right] = 2\left[f''(\alpha)\cdot 2x^2 + f'(\alpha)\right]$$

$$\frac{\partial P}{\partial y} = f'(\alpha)\cdot\frac{\partial \alpha}{\partial y} = f'(\alpha)\cdot(-2y)$$

$$\frac{\partial^2 P}{\partial y^2} = -2\left[f''(\alpha)\cdot(-2y)\cdot y + f'(\alpha)\right] = 2\left[f''(\alpha)\cdot 2y^2 - f'(\alpha)\right]$$

$$\Delta P = 0 \Rightarrow 2\left[f''(\alpha)2x^2 + f'(\alpha)\right] + 2\left[f''(\alpha)\cdot 2y^2 - f'(\alpha)\right] = 0$$

$$\Rightarrow 4f''(\alpha)(x^2+y^2) + 2f'(\alpha) - 2f'(\alpha) = 0 \Rightarrow$$

$$\Rightarrow 4(x^2+y^2)f''(\alpha) = 0 \Rightarrow f''(\alpha) = 0 \Rightarrow f'(\alpha) = k,\ k\in\mathbb{R}$$

$$f(\alpha) = k\alpha + a \quad \text{ш } k, a\in\mathbb{R}$$

$$P(x,y) = k(x^2-y^2) + a \qquad k, a\in\mathbb{R}$$

$$\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = 2kx \xrightarrow{\int dy} Q(x,y) = \int 2kx\,dy \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = -2ky \end{cases}$$

$$Q(x,y) = 2xky + c(x)$$

$$\frac{\partial Q}{\partial x} = 2ky + c'(x) = 2ky \Rightarrow c'(x) = 0 \Rightarrow c(x) = b,\ b\in\mathbb{R}$$

$$Q(x,y) = 2xyk + b \qquad k, b\in\mathbb{R}$$

$$f = P + iQ = k(x^2-y^2) + a + i(2xyk+b)$$

$$f = k(x^2 - y^2 + 2ixy) + a + ib = k(x+iy)^2 + a + ib$$

$$f(z) = kz^2 + a + ib \qquad k, a, b\in\mathbb{R}$$